José Junior Miksza

SENIOR SOFTWARE ENGINEER | FULLSTACK ENGINEERING | AWS | APPLIED AI

Campinas, São Paulo, Brazil
Open to remote, hybrid, and on-site | Open to international relocation

Email: junior.miksza8@pm.me | Phone: +55 19 99757-9762
LinkedIn: https://www.linkedin.com/in/jos%C3%A9-junior-miksza-4766a8199 | GitHub: https://github.com/JuniorMiksza8

PROFESSIONAL SUMMARY

Senior Software Engineer with 7+ years of experience designing and shipping high-throughput, cloud-native systems across fintech, e-commerce, and tax compliance. Fullstack expertise spanning Java/Spring and Node.js/TypeScript microservices, React/Next.js frontends, and AWS infrastructure, with a growing focus on applied AI — LLM-powered workflows, embeddings, and low-latency search engines. Delivered measurable impact at scale: improved LLM workflow success rates from 78% to 92%, cut settlement batch processing time by 40%, and sustained 1M+ daily transactions with resilient, event-driven architectures. Full lifecycle owner from system design to production, with a track record of mentoring engineers and raising the bar through architecture reviews.

SKILLS

- **Backend Engineering:** Java, Spring, Spring Batch, Apache Lucene, Node.js, TypeScript, API development, microservices, third-party integrations
- **Cloud Architecture:** AWS, cloud-native services, distributed systems, IAM, networking, event-driven design, cost-aware scalability
- **Financial Systems:** Payment settlement, sub-acquirer settlement, receivables anticipation (CERC), Pix, transaction management, dispute workflows, fixed-width banking files, financial reporting
- **Data & Performance:** SQL, NoSQL, caching, queues, database optimization, high-volume processing
- **Applied AI:** LLM-powered applications, text embeddings, vector similarity search, intelligent automation
- **Frontend:** React, Next.js, SSR, Material UI, design systems, administrative dashboards
- **Engineering Practices:** Automated testing, CI/CD, observability, incident analysis, architecture reviews, technical mentoring

PROFESSIONAL EXPERIENCE

Avalara - Software Engineer III

Mar 2025 - Present

Develop scalable, LLM-powered systems for a greenfield initiative supporting Avalara's AI-first strategy in tax compliance automation.

- Design backend components for intelligent automation, collaborating with cross-functional teams to translate business requirements into maintainable software.
- Owned technical delivery of 3 production services, from API and data-flow design through implementation, testing, deployment, and monitoring.
- Improved an LLM workflow's task success rate from 78% to 92% across a 300-case evaluation set by introducing structured outputs, validation, and failure-handling mechanisms.

- Reduced average inference cost per completed workflow by 25% through model selection, context optimization, and caching, while preserving agreed quality thresholds.
- Established regression tests and operational dashboards for critical AI workflows, enabling the team to detect quality degradation before production releases.

Key Projects

Denied Party Screening Engine

- Architected a fault-tolerant ingestion platform that continuously synchronizes 30+ global sanctions lists (OFAC, EU, UN, UK HMT, etc.) covering 2M+ entities, using chunk-oriented Spring Batch jobs with checkpoint/restart, idempotent upserts, and delta detection, orchestrated by Kubernetes CronJobs with raw list snapshots versioned in S3.
- Achieved sub-20ms p95 lookup latency per screening by indexing normalized entity data (names, aliases, addresses) in Apache Lucene with tuned analyzers for fuzzy and phonetic matching.
- Reduced false-positive review workload by 35% through scoring and threshold tuning across name-matching strategies.

Cross-Rulings Semantic Search

- Built a resilient ingestion pipeline processing 500K+ CBP and EBTI ruling documents, with Spring Batch jobs orchestrated via Kubernetes CronJobs, raw documents versioned in S3, and normalized, deduplicated records persisted in MongoDB with full reprocessing support.
- Designed a multi-language semantic matching service that embeds rulings and analyst queries into a shared vector space, using cosine similarity scoring with metadata pre-filtering to deliver sub-second retrieval across 10+ languages, eliminating the recall limitations of keyword-based search.
- Cut average ruling research time from hours to minutes by surfacing the most relevant precedents through semantic ranking with tunable relevance thresholds.

Vega Solutions - Software Developer

Aug 2022 - May 2025 | Remote

Developed Java/Spring applications for financial institutions, supporting transaction management, disputes, and settlement workflows involving Mastercard, Visa, and Elo.

- Built services for fixed-width banking file processing, financial reporting, and API integrations, translating institution-specific requirements into backend processing workflows.
- Reduced settlement batch processing time by 40%, from 50 to 30 minutes, through profiling, database query optimization, and changes to transaction processing routines.
- Supported workloads exceeding 1 million transaction records per day, implementing validation and reconciliation checks to identify incomplete or inconsistent settlement data.
- Reduced manual investigation time by 30% by introducing structured logs, transaction-level traceability, and actionable processing error reports.
- Strengthened retry safety and duplicate-processing protection, reducing incidents caused by repeated file submissions or API retries.
- Reviewed critical backend changes and mentored 2 developers on service design, testing, and maintainability.

Key Projects

Receivables Anticipation Platform (CERC-integrated)

- Built a receivables anticipation platform integrated with CERC (Brazil's central receivables registry), automating registration, querying, and locking of receivables to secure anticipation operations, using Java/Spring Boot with idempotent API integrations and retry-safe state machines.
- Processed thousands of anticipation requests daily with full auditability, reducing manual registry reconciliation effort by 60% through automated CERC file exchange and status tracking.

Sub-Acquirer Settlement Engine

- Designed a multi-tenant settlement engine for a sub-acquirer, computing fee splits, commissions, and payout schedules across merchants, with automated reconciliation against card network and banking files.
- Improved settlement accuracy and traceability by introducing transaction-level reconciliation reports and duplicate-processing guards, supporting 1M+ daily transaction records.

MyPharma - Software Developer

Jul 2020 - Oct 2022

Developed backend services and frontend applications for a pharmacy e-commerce platform, covering authentication, store management, integrations, and payment workflows.

- Built Node.js/TypeScript microservices backed by NoSQL databases and React/Material UI dashboards for pharmacy owners and internal operations.
- Reduced p95 API response time by 35%, from 800 ms to 520 ms, through targeted caching, database index improvements, and query optimization.
- Improved background processing throughput by 50% by moving long-running integration tasks to queues and tuning worker concurrency.
- Integrated payment gateways and developed operational tools supporting store administration and customer purchase flows.
- Increased successful payment completion from 94% to 97% through improved error handling, integration diagnostics, and recovery workflows.
- Optimized SEO for dynamically generated pages and tuned application performance through caching, queues, and database improvements.

Key Projects

Marketplace Re-Platforming

- Led the re-platforming of a legacy monolith into Node.js/TypeScript microservices handling payment management and settlement, including Pix integration and multiple payment gateway providers with webhook-based reconciliation.
- Rebuilt the pharmacy marketplace storefront with Next.js, server-side rendering, and a shared design system, improving organic traffic by 40% through SEO-optimized dynamic pages and better Core Web Vitals.

Backoffice & Cloud Migration

- Built a backoffice platform for online store management (catalog, orders, payments, operations) with role-based access control (IAM) and a redesigned MongoDB data model supporting multi-store tenancy.
- Drove cloud adoption by migrating workloads to AWS (containerized services, managed databases, S3/CloudFront for assets), improving deployment frequency and environment consistency.

Pharmainox - IT Apprentice

Apr 2019 - Jun 2020

- Developed an internal resource management system and contributed to a vehicle control application supporting administrative operations.

- Reduced time spent on recurring administrative tasks by 6 hours per week by replacing spreadsheet-based routines with application workflows.
- Worked directly with internal users to clarify requirements, validate functionality, and improve usability.

EDUCATION

MBA, Financial Management, Auditing and Compliance
UniFAJ | 2023 - 2024

Undergraduate Degree, Software Development
Universidade Paulista | 2020 - 2022

Technical Qualification, Software Development
SENAI | 2019 - 2020

CERTIFICATIONS

- **AWS Certified Cloud Practitioner**

LANGUAGES

- **English:** EF SET - Overall C2 Proficient | Mar 2025
- **Portuguese:** Native

KEYWORDS

- **Java:** Java, JVM, Maven, Gradle, JUnit, Mockito, JDBC, JPA, Hibernate
- **Spring:** Spring Boot, Spring Framework, Spring MVC, Spring WebFlux, Spring Security, Spring Data, Spring Cloud, Spring Batch, Spring Actuator
- **AWS:** EC2, ECS, EKS, Lambda, S3, RDS, DynamoDB, SQS, SNS, EventBridge, API Gateway, CloudWatch, IAM, VPC, CloudFront, ElastiCache, ALB, Route 53, Secrets Manager, CloudFormation
- **React:** React, ReactJS, Hooks, Redux, React Router, Material UI, MUI, Next.js, SSR, SEO
- **Node.js & TypeScript:** Node.js, Express, NestJS, TypeScript, npm, REST APIs, GraphQL, Pix, payment gateways, CERC
- **AI & Search:** text embeddings, vector similarity search, Apache Lucene, LLM-powered applications
- **Kubernetes & Helm:** Kubernetes, K8s, Helm, Helm charts, kubectl, Ingress, ConfigMaps, Secrets, Deployments, Services, HPA, RBAC
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, Redis, DynamoDB, Elasticsearch, SQL, NoSQL
- **Queues & Messaging:** SQS, SNS, RabbitMQ, Apache Kafka, Redis, JMS, event-driven architecture
- **CI/CD & Platform:** Docker, Git, GitHub Actions, Jenkins, Terraform, Linux, CI/CD, microservices